

BENJAMIN ADJEPONG

US Citizen – Eligible for Clearance – Willing to Relocate

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Objective

Master of Engineering candidate in Software and Machine Intelligence seeking full-time software engineering roles. Leveraging enterprise experience in AI/ML frameworks, test automation, and cross-platform UI development.

Education

Virginia Tech

Expected May 2026

Master of Engineering in Computer Engineering, Software and Machine Intelligence — GPA: 3.90

Blacksburg, VA

Virginia Tech

August 2021 – May 2025

Bachelor of Science in Computer Engineering, Machine Learning — GPA: 3.64

Blacksburg, VA

Relevant Coursework: Applications of Machine Learning, Artificial Intelligence, Brain-Inspired Computer Architecture, Computer Vision, Deep Learning, Real-Time Systems, Trustworthy Machine Learning, Video Game Design & Engineering

Skills

Languages: Python, PowerShell, KQL, C/C++, C#, JavaScript, R, Verilog/VHDL

Frameworks & Tools: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, Pandas, NumPy, Unity, Git/GitHub

Hardware & Platforms: Raspberry Pi 5, Nvidia Jetson Nano, Meta Quest 3, TI MSP432, Arduino

Experience

Automation Creations, Inc.

September 2025 – March 2026

Software Engineer – UI Design | C++, Qt/QML

Blacksburg, VA

- Developed interactive screens, data tables, and reusable UI components for a data-driven customization application
- Engineered dynamic front-end architectures by binding complex backend data streams to responsive user interfaces
- Operated as a versatile contractor, resolving diverse Agile backlog tickets from UI enhancements to feature additions

Microsoft

May 2025 – August 2025

Software for Hardware Engineer Intern – Test Automation & Reporting

Redmond, WA

- Automated test execution pipeline using PowerShell, accelerating framework deployment and reducing manual testing
- Engineered a scalable data integration framework to continuously monitor hardware via 1000's of daily telemetry points
- Built a centralized Power BI dashboard consolidating test data, improving cross-functional bug resolution speed

Microsoft

May 2024 – August 2024

Software for Hardware Engineer Intern – Log Based Anomaly Detection

Redmond, WA

- Developed a scalable AI/ML anomaly detection framework to assess hardware operational health via driver device logs
- Built an automated data pipeline to clean, parse, and transform raw telemetry into model-ready datasets
- Trained an unsupervised machine learning model with 90%+ precision for proactive hardware failure mitigation

Collins Aerospace

June 2023 – August 2023

SEPP Mission Systems Software Engineer Intern – Station Keeping Playbook (SKP)

Sterling, VA

- Developed operational testing platform for unmanned aerial systems on ground robots, reducing testing costs by 90%
- Tested autonomous capabilities for UASs by integrating advanced autonomous flight software onto ground robots
- Implemented Python libraries to process GPS & IMU telemetry data, commanding motors controls via SKP protocols

Projects

Mixed Reality Radio Visualization | Unity, C#, Python, Meta Quest 3, Raspberry Pi 5 February 2026 – May 2026

- Engineered a mixed reality (AR/VR) tool to visualize high fidelity broadband spectrometer as isometric waterfall
- Developed a cross-platform data pipeline to ingest, transmit, and visualize raw I/Q data within VR/AR environment
- Implemented immersive user interface for real-time spectral analysis, enabling dynamic frequency tuning and monitoring

ML Classification of Electrochemical Fingerprints of Milk | Python, scikit-learn

August 2024 – May 2025

- Architected a normalization pipeline and engineered frequency-band features from 61-dimensional EIS spectra
- Implemented Leave-One-Cow-Out GroupKFold cross-validation to eliminate data leakage and ensure generalization
- Deployed a k-NN classification model achieving 90.8% accuracy and 0.888 F1-score in detecting milk perturbations

Facial Attribute Detection | CV, Python

September 2024 – December 2024

- Architected a real-time computer vision pipeline utilizing MTCNN (P-Net, R-Net, O-Net) to extract 20 facial attributes
- Optimized Multi-CNN network layers and hyperparameters to balance live inference speed with detection accuracy
- Engineered a live video processing system that tracks facial bounding boxes and computes attribute probabilities